Code No: 118AA **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2021
ADHOC AND SENSOR NETWORKS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) What is a Manet? In what way it is different from conventional Mobile Networks and Wireless Networks?
b) Distinguish the following design issues of a Manet:
i) Interoperation Vs Interoperation with Internet
ii) Energy Conservation Vs Node Cooperation. [7+8]

2.a) Compare DSR and DSDV algorithms using the criteria.
b) Illustrate the working of AODV algorithm. What are its pros and cons? [7+8]

3.a) Explain the problem of redundant packet retransmission in Manet broadcast with an example.
b) Write the algorithmic steps for MPR broadcast algorithm. Illustrate with an example. [7+8]

4.a) Explain the two Brach Reconstruction (BR) modules of AMRIS, a tree-based multicast algorithm for Manets.
b) Distinguish the following terms:
i) Sensor ii) Sensor Node
iii) Sensor Network iv) Mobile Robots. [7+8]

5.a) Explain why routing in Wireless Sensor Networks is different and challenging compared to conventional networks.
b) Illustrate how a CH is decided in LEACH algorithm with a numeric example. [7+8]

6.a) What are the security challenges in ad hoc and sensor networks?
b) Explain how Diffie-Hellman algorithm is used for key management in Ad hoc and Sensor networks. [7+8]

7.a) Write about Sensor Network Programming Challenges.
b) List the features of nesC programming language. Explain any two. [7+8]

8.a) Compare the two: TinyGALS and TOSSIM.
b) Write about network simulator version 2 (NS2). [7+8]

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